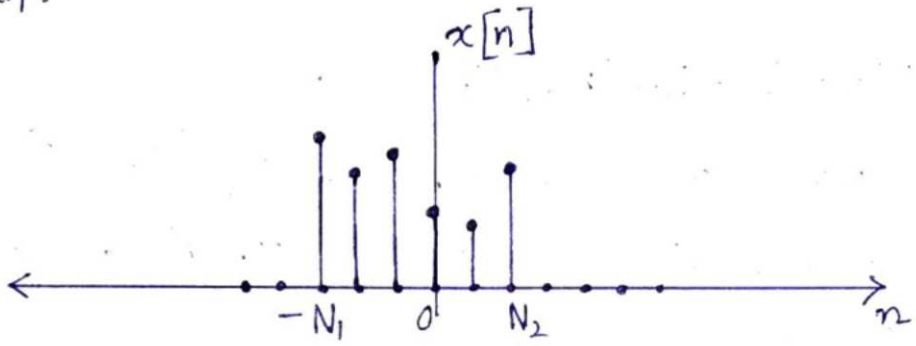
Discrete Time Fourier Transform

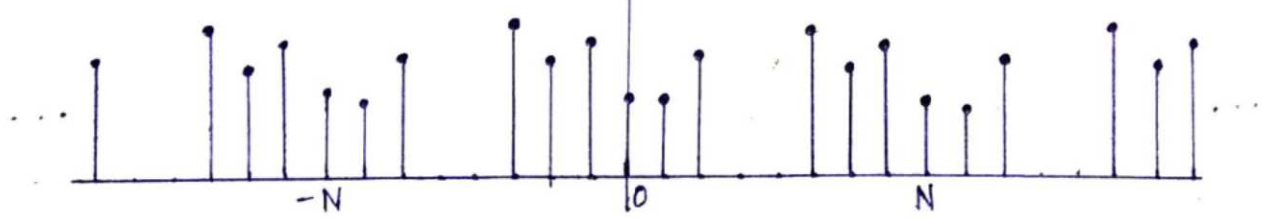
Similar to continuous time aperiodic signals, we look into discrete time aperiodic signals and their frequency domain representations in this Module.

Consider a general sequence $x[n]$ that is of finite duration, i.e. $x[n] = 0$ for $n \leqslant N_2$ and $n \geqslant -N_1$.



Now, we do a periodic extension of $x[n]$.
(Similar to the continuous time case)

$$x_N[n] := \sum_{k=-\infty}^{\infty} x[n+kN]$$



$$\therefore \lim_{N\to\infty} x_N[n] = \lim_{N\to\infty} \sum_{k=-\infty}^{\infty} x[n+kN] = x[n]$$

The Fourier series representation of $x_N[n]$ is

$$x_N[n] = \sum_{k=\langle N \rangle} a_k e^{jk(2\pi/N)n} \longrightarrow (1)$$

where $$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x_N[n] e^{-jk(2\pi/N)n} \longrightarrow (2)$$

Since $x[n] = x_N[n]$ over a period that includes the interval $-N_1 \leqslant n \leqslant N_2$; it is convenient to choose the interval of summation in equation (2) to include this interval, so that $x_N[n]$ can be replaced by $x[n]$ in the summation.

$$\therefore \quad a_k = \frac{1}{N} \sum_{n=-N_1}^{N_2} x[n] e^{-jk(2\pi/N)n}$$

$$= \frac{1}{N} \sum_{n=-\infty}^{\infty} x[n] e^{-jk(2\pi/N)n} \longrightarrow (3)$$

This can be done because $x[n]$ is 0 outside $[-N_1, N_2]$.

Defining

$$X(e^{j\omega}) := \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n} \longrightarrow (4)$$

We can write from equation (3),

$$a_k = \frac{1}{N} X(e^{jk\omega_0}), \text{ where } \omega_0 = 2\pi/N.$$

$$\therefore (1) \Rightarrow x_N[n] = \sum_{k=\langle N\rangle} \frac{1}{N} X(e^{jk\omega_0})\, e^{jk\omega_0 n}$$

$$= \frac{\omega_0}{2\pi} \sum_{k=\langle N\rangle} X(e^{jk\omega_0})\, e^{jk\omega_0 n}$$

$$\left(\because \frac{2\pi}{N} = \omega_0 \Rightarrow \frac{1}{N} = \frac{\omega_0}{2\pi}\right)$$

Similar to the continuous case we will have

$$\lim_{N\to\infty} x_N[n] = \lim_{N\to\infty} \frac{1}{2\pi} \sum_{k=\langle N\rangle} X(e^{jk\omega_0})\, e^{jk\omega_0 n}\, \omega_0$$

$$\Rightarrow \boxed{x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega}$$

$\because$ summation is carried over N consecutive intervals of width $\omega_0 = 2\pi/N$, the total interval of integration will be 2π.

∴ The Fourier transform equation for discrete-time signals is —

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$ Analysis equation

→ (A)

and the inverse Fourier transform equation is

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$ Synthesis equation.

→ (B)

This indicates how an aperiodic discrete time signal can be thought of as a linear combination of complex exponentials. The synthesis equation is a representation of $x[n]$ as a linear combination of complex exponentials infinitesimally close in frequency and with amplitudes $X(e^{j\omega})(d\omega/2\pi)$. Therefore, $X(e^{j\omega})$ is called the spectrum of $x[n]$.

Let us compare this with DT-Fourier series.

Periodic signal $\left(\omega_0 = \frac{2\pi}{N}\right)$	Aperiodic signal
$\downarrow$	$\downarrow$
DT-Fourier Series	DT-Fourier Transform
* Synthesis equation $x[n] = \sum_{k=\langle N\rangle} a_k e^{j\left(\frac{2\pi}{N}\right)kn}$ → Finite-sum	* Synthesis equation $x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$ → Integration
* Analysis equation $a_k = \sum_{n=\langle N\rangle} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn}$	* Analysis equation $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$
* Periodic signals can be represented as sum of complex exponentials admitted only at a discrete set of harmonically related frequencies.	* Aperiodic signals can be represented as sum of exponentials ($\because$ integral is sum) but these exponentials occur at continuum of frequencies.
* The set a_k is called the spectrum. This is a discrete-set and periodic with period N.	* $X(e^{j\omega})$ is called the spectrum. This is not a discrete-set and is periodic with period 2π.

Apart from the fact that CTFT is for continuous time and DTFT is for discrete time, there are similarities between the two as was evident by the way we did the derivation. However there are two major differences —

(1) In CTFT, the spectrum $X(j\omega)$ was not periodic. But in DTFT, the spectrum $X(e^{j\omega})$ is periodic with period 2π.

$$X\left(e^{j(\omega+2\pi)}\right) = \sum_{n=-\infty}^{\infty} x[n] e^{-j(\omega+2\pi)n}$$

$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \cdot \underbrace{e^{-j2\pi n}}_{\to 1}$$

$$= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = X(e^{j\omega})$$

$\therefore$ The spectrum $X(e^{j\omega})$ is periodic with period 2π.

② The interval of integration is the synthesis equation is finite. In DTFT the integration is over 2π interval.

<u>Example 1:</u> $x[n] = \alpha^n u[n], \quad |\alpha| < 1$

$$\therefore X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \alpha^n u[n] e^{-j\omega n}$$

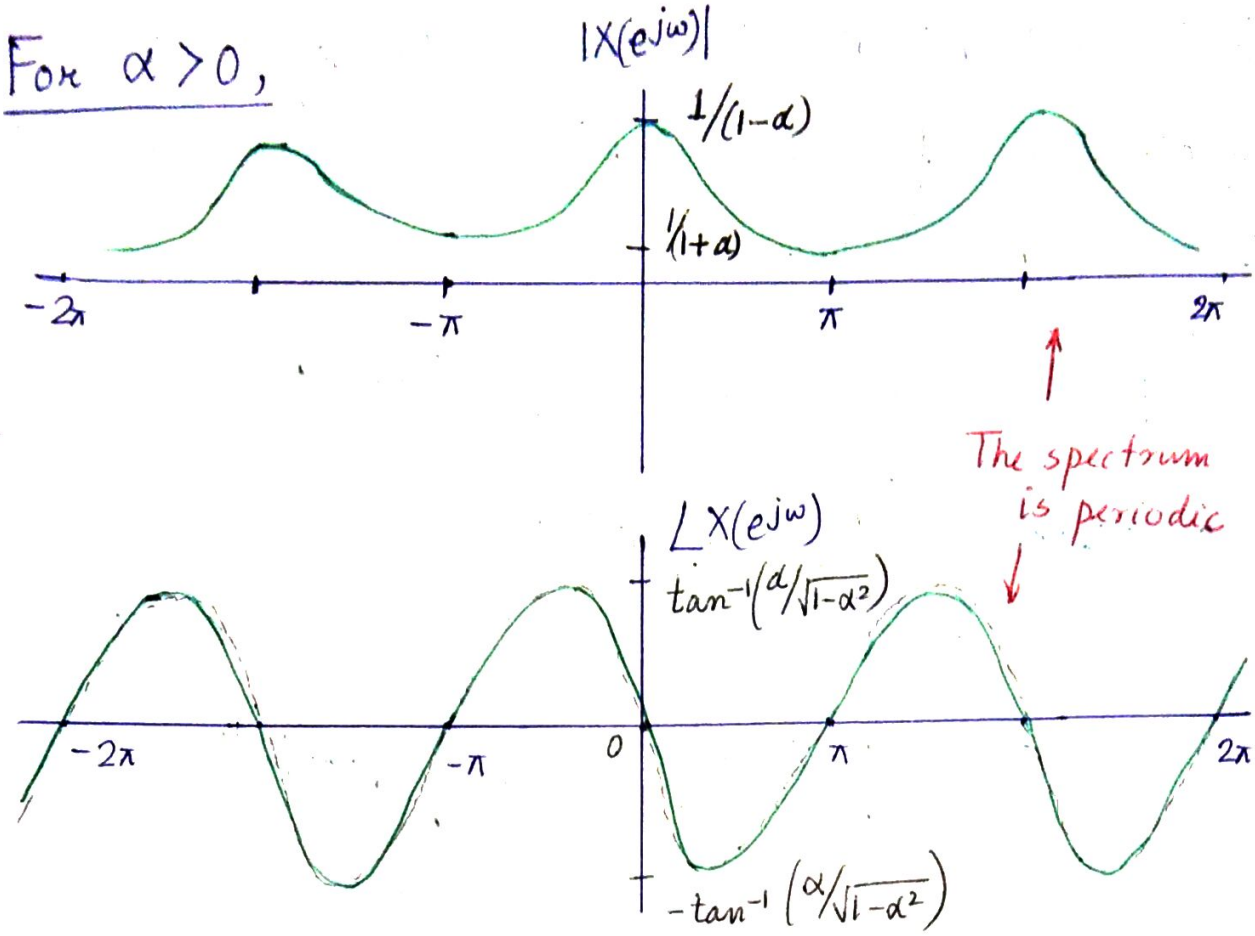
$$= \sum_{n=0}^{\infty} \alpha^n e^{-j\omega n} = \sum_{n=0}^{\infty} (\alpha e^{-j\omega})^n$$

$$= \frac{1}{1 - \alpha e^{-j\omega}}$$

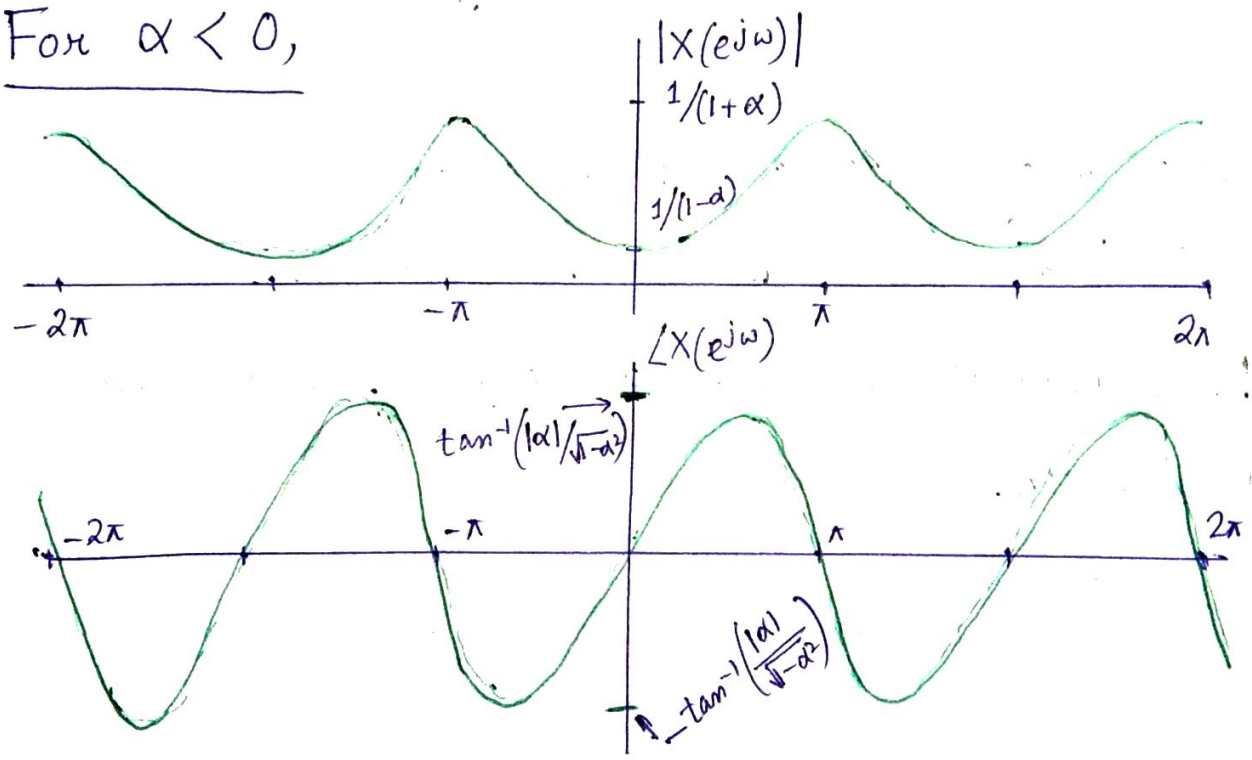
The magnitude and phase ~~response~~ of $X(e^{j\omega})$ are shown in the next figures.

Note that the functions are periodic in ω with period 2π.

For $\alpha > 0$,



For $\alpha < 0$,



What is peculiar about all the analysis we did is that although we are dealing with <u>aperiodic</u> discrete signals, the spectrum is <u>periodic</u> with period 2π.

Again, remember although we derived the expression DTFT only for finite-support aperiodic DT signals, the equations (A) & (B) are valid for an extremely broad class of signals with infinite duration. In this case however we need to ask the question of convergence of the infinite sum (notice eq^n (A) is an infinite sum). This was not a problem in DTFS as we had finite sum.

<u>Convergence issues</u>:

The conditions on $x[n]$ that guarantee convergence are similar to that for the CTFT.

In particular, equation (A) converges if $x[n]$ is absolutely summable, i.e.,

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

or if the sequence has finite energy

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 < \infty.$$

This are <u>sufficiency</u> conditions only.

Equation (B) doesnot have convergence issues since it is a finite-interval integral. There we would expect not to see any behavior like Gibbs phenomenon in evaluating the discrete-time Fourier transform synthesis equation.

e.g. Take the signal $x[n] = \delta[n]$.

$$\therefore X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n} = 1 .$$

Now, let us try to use the synthesis equation again.

$$\hat{x}[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{0}^{2\pi} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \left[\frac{e^{j\omega n}}{jn} \right]_{0}^{2\pi} = \frac{1}{2\pi} \left[\frac{e^{j2\pi n}}{jn} - \frac{1}{jn} \right]$$

$$\therefore \hat{x}[n] = \frac{1}{j2\pi n} \left(e^{j2\pi n} - 1 \right)$$

$$\therefore \lim_{n \to 0} \hat{x}[n] = \lim_{n \to 0} \frac{1}{j2\pi n} \left(e^{j2\pi n} - 1 \right) = 1$$

(Use L'Hospital's rule)

and for $n \neq 0$, $\hat{x}[n] = 0$

$$\therefore \hat{x}[n] = \delta[n].$$

#

Thus, we get back the actual signal. Recall that discrete-signals can be written as

sum of shifted unit-samples.
Hence, using linearity we can infer that we won't have Gibbs' phenomenon here.

Fourier Transform for Periodic signals

The question that we ask here again is what happens if we take Fourier transform of a periodic signal.

Check that in module-8a we found that the CTFT of

$$x(t) = e^{j\omega_0 t} \quad \text{is} \quad X(j\omega) = 2\pi\,\delta(\omega - \omega_0).$$

Similarly, we try to find the DTFT of $x[n] = e^{j\omega_0 n}$.

We mighted be tempted to think that $X(e^{j\omega}) = 2\pi\,\delta(\omega - \omega_0)$ — But this is not correct.

This is not correct because for DTFT we know that the spectrum is periodic with a period of 2π.

This means we need to repeat $2\pi\delta(\omega-\omega_0)$ after every 2π interval. Hence, we guess that the spectrum of $e^{j\omega_0 n}$ is

$$X(e^{j\omega}) = \sum_{l=-\infty}^{\infty} 2\pi\,\delta(\omega-\omega_0-2\pi l) \longrightarrow (5)$$

This is to sum the shifted impulses together

This takes into account the periodicity.

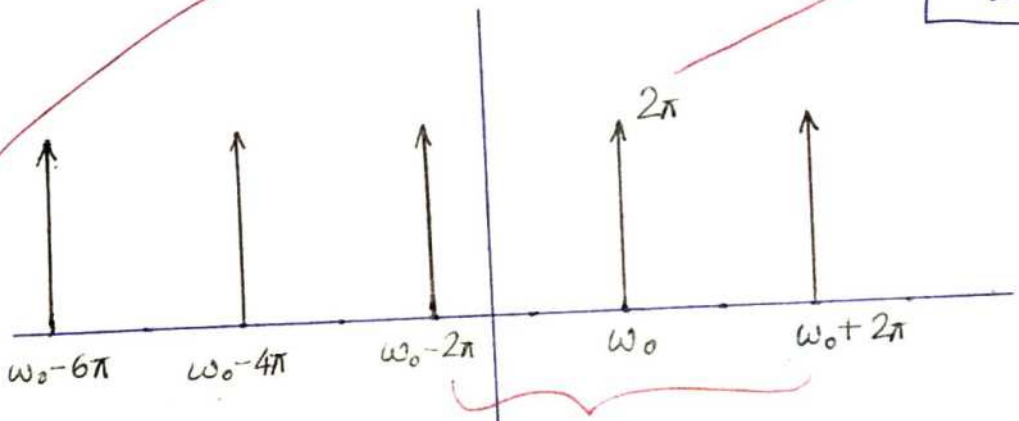
Let us verify whether $X(e^{j\omega})$ in equation (5) is indeed the spectrum of $x[n] = e^{j\omega_0 n}$. Using the inverse DTFT —

$$\hat{x}[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{-j\omega n}\, d\omega$$

$$\therefore \hat{x}[n] = \frac{1}{2\pi} \int_{2\pi} \underbrace{\sum_{l=-\infty}^{\infty} 2\pi\, \delta(\omega - \omega_0 - 2\pi l)}_{\text{Over a period } 2\pi \text{ we will just have one impulse of the summands.}} e^{+j\omega n}\, d\omega$$

$$\therefore \hat{x}[n] = \frac{1}{2\pi} \int 2\pi\, \delta(\omega - \omega_0)\, e^{+j\omega n}\, d\omega$$

We are chosing the interval of integration to have the impulse at ω_0



We are chosing this 2π - width interval for the integration. Hence we have $2\pi\, \delta(\omega - \omega_0)$

$$\therefore \hat{x}[n] = \int \delta(\omega - \omega_0)\, e^{+j\omega n}\, d\omega = e^{+j\omega_0 n} = x[n]$$

$\therefore$ Our guess is correct.

$$\therefore \quad e^{j\omega_0 n} \xleftrightarrow{\mathcal{F}} \sum_{l=-\infty}^{\infty} 2\pi\, \delta(\omega - \omega_0 - 2\pi l)$$

Now we consider a periodic DT signal.

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{j(2\pi/N)kn}$$

This is the Fourier series representation of the periodic signal

$$\therefore x[n] = a_0 + a_1 e^{j\omega_0 n} + a_2 e^{j2\omega_0 n} + \ldots + a_{N-1} e^{j(N-1)\omega_0 n}$$

Note that

$$a_0 \xleftrightarrow{\mathcal{F}} \sum_{l=-\infty}^{\infty} 2\pi a_0 \delta(\omega - 2\pi l)$$

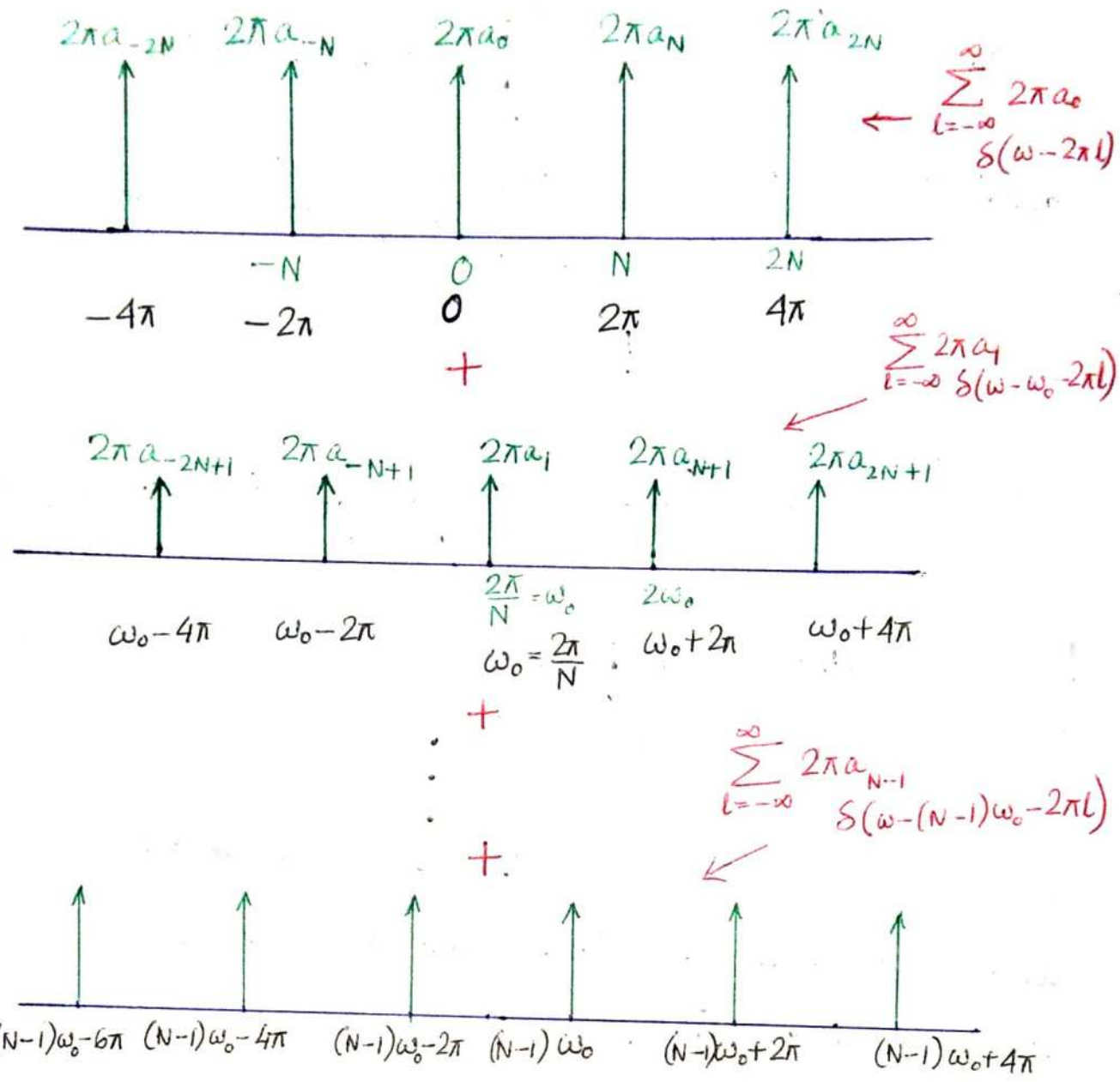
$$a_1 e^{j\omega_0 n} \xleftrightarrow{\mathcal{F}} \sum_{l=-\infty}^{\infty} 2\pi a_1 \delta(\omega - \omega_0 - 2\pi l)$$

$$\vdots$$

$$a_{N-1} e^{j(N-1)\omega_0 n} \xleftrightarrow{\mathcal{F}} \sum_{l=-\infty}^{\infty} 2\pi a_{N-1} \delta(\omega - (N-1)\omega_0 - 2\pi l)$$

Further, also recall from the analysis in module 6a that $a_k = a_{k+N}$.

We graphically represent the spectrum next and add them up. This will give us the spectrum for $x[n]$.


$2\pi a_{-2N}$
$2\pi a_{-N}$
$2\pi a_0$
$2\pi a_N$
$2\pi a_{2N}$
$\sum_{l=-\infty}^{\infty} 2\pi a_0 \, \delta(\omega - 2\pi l)$
$-N$
0
N
$2N$
-4π
-2π
0
2π
4π
$+$
$\sum_{l=-\infty}^{\infty} 2\pi a_1 \, \delta(\omega - \omega_0 - 2\pi l)$
$2\pi a_{-2N+1}$
$2\pi a_{-N+1}$
$2\pi a_1$
$2\pi a_{N+1}$
$2\pi a_{2N+1}$
$\frac{2\pi}{N} = \omega_0$
$2\omega_0$
$\omega_0 - 4\pi$
$\omega_0 - 2\pi$
$\omega_0 = \frac{2\pi}{N}$
$\omega_0 + 2\pi$
$\omega_0 + 4\pi$
$+$
$\sum_{l=-\infty}^{\infty} 2\pi a_{N-1} \, \delta(\omega - (N-1)\omega_0 - 2\pi l)$
$+$
$(N-1)\omega_0 - 6\pi$
$(N-1)\omega_0 - 4\pi$
$(N-1)\omega_0 - 2\pi$
$(N-1)\omega_0$
$(N-1)\omega_0 + 2\pi$
$(N-1)\omega_0 + 4\pi$
$\|$
$2\pi a_0$
$2\pi a_{N-1}$
$2\pi a_1$

From the figure it is clear that

$$\boxed{X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \, \delta(\omega - k\omega_0)}, \quad \omega_0 = \frac{2\pi}{N} \longrightarrow (6)$$

Thus, co-efficients of Fourier series and the spectrum $X(e^{j\omega})$ of a periodic signal are related by equation (6).

Similar to the continuous-time periodic signals (see Module-8a page 19), the Fourier transform of a periodic signal with Fourier coefficients $\{a_k\}$ can be interpretated as a train of impulses occuring at the harmonically related frequencies and for which the area of the impulse at the k-th harmonic frequency $k\omega_0$ is 2π times the k-th Fourier series co-efficient a_k.

(Note $\delta(\omega - k\omega_0)$ in eq^n (6) is impulse not unit sample $\delta[n]$)

Example: Consider the example in Module - 6a page 8.

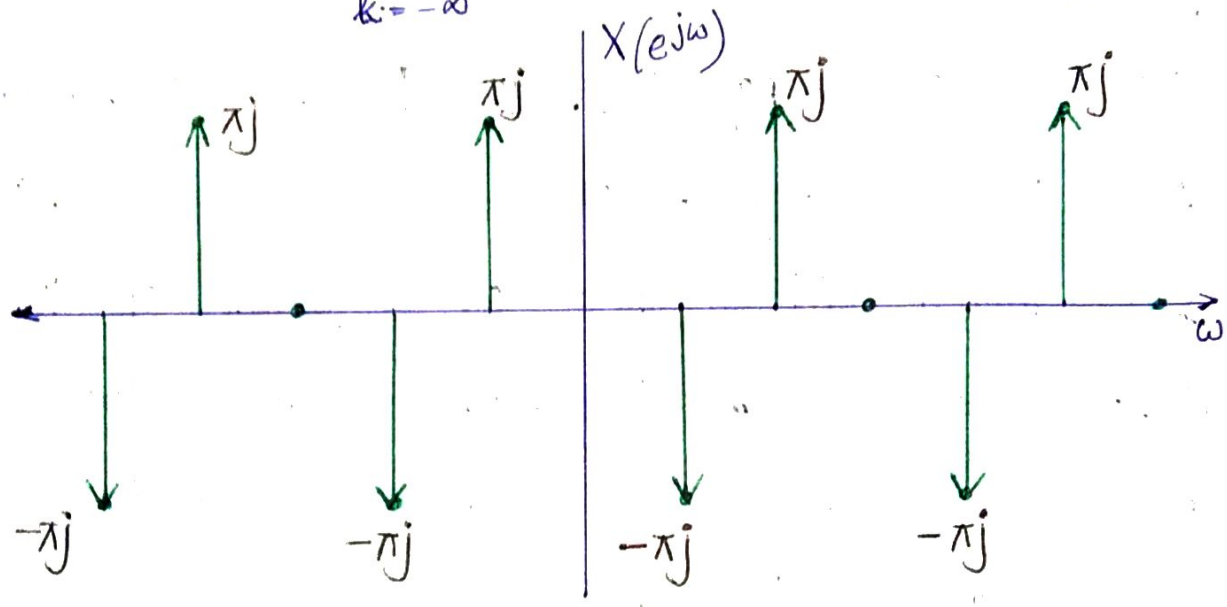
$$x[n] = \sin\left(\frac{2\pi}{N}\right) n \quad , \quad N = 3$$

$$\therefore \quad a_0 = 0 \; , \quad a_1 = -1/2j \; , \quad a_2 = 1/2j$$

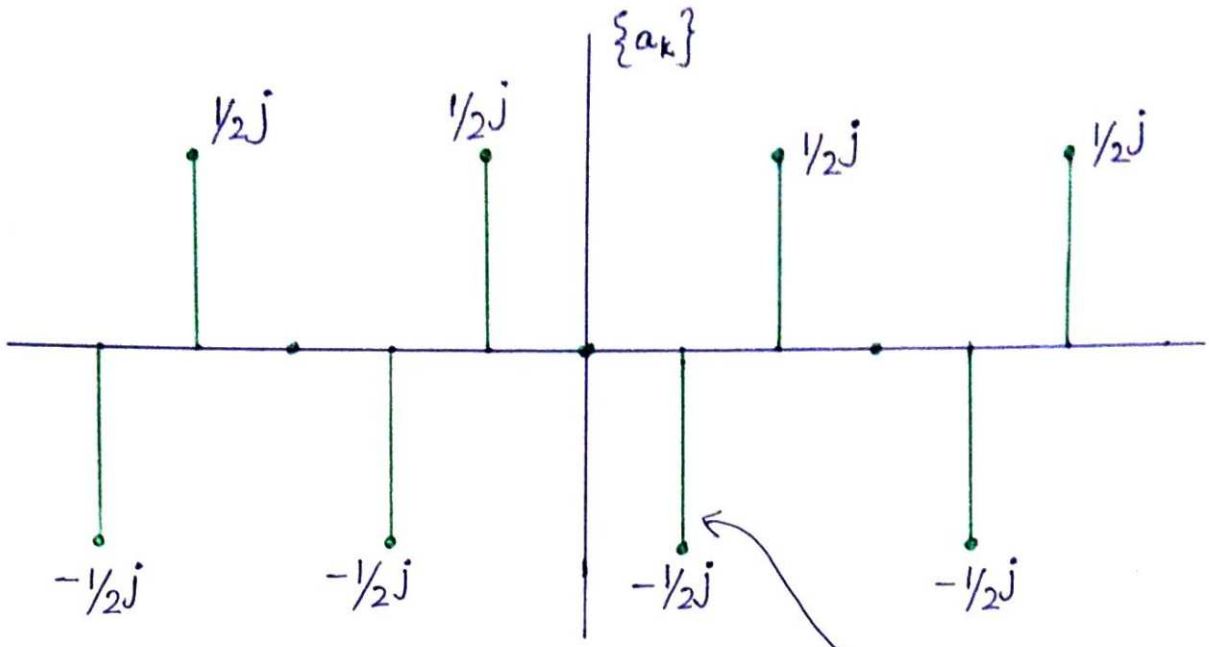
From our previous discussion,

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} 2\pi a_k \, \delta\left(\omega - k\frac{2\pi}{3}\right)$$

This is not unit sample but an impulse with area 1.

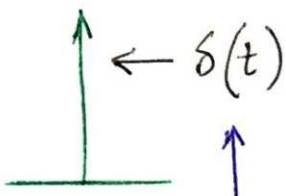


But notice when we draw the Fourier-series coefficients for the same signal we use unit samples.

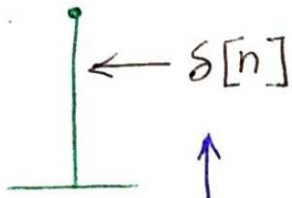


This is a sample of magnitude $+1/2$, i.e. $-\frac{1}{2}j\,\delta[n-1]$

Notice that I am not using the arrows while drawing the spectrum



Here talk about unit area — continuous case



Here we talk about unit magnitude — discrete case

[The same idea is true for the continuous case also — check example in page 21 of module 8a]